

JAMAL AL JAZEER F

Biomedical Engineer | Healthcare Equipment Service Engineer | AI/ML Minor

Dubai, UAE | jamalaljazeer@gmail.com | +971 559723786 | Available Immediately | Indian National

Portfolio: jamalaljazeer.vercel.app/ | **LinkedIn Profile:** www.linkedin.com/in/jamal-al-jazeer-1225832b2

Languages: English, Tamil, Malayalam

PROFESSIONAL SUMMARY

Motivated Biomedical Engineering graduate (B.E., AI/ML Minor, CGPA 8.37/10) with hands-on clinical and field service experience in hospital and medical equipment environments. Skilled in preventive maintenance (PPM), corrective maintenance (CMM), calibration, installation, commissioning, and after-sales technical support for patient monitors, ventilators, infusion pumps, ECG machines, and diagnostic imaging systems. Holds a registered Design Patent (IP India, 2026) and two co-authored research publications. Immediately available for relocation to Dubai, UAE for Biomedical Technician, Biomedical Engineer, or Field Service Engineer roles.

CORE COMPETENCIES

Preventive Maintenance (PPM) | Corrective Maintenance (CMM) | Equipment Calibration & Validation
Medical Equipment Installation & Commissioning | After-Sales Technical Support | Field Service
Patient Monitor, Ventilator & Infusion Pump Servicing | Diagnostic Imaging & Radiology Equipment Support
Biomedical Signal Processing (ECG, EMG, PPG, EEG) | IoT-Based Healthcare Device Integration
Technical Documentation & Maintenance Reporting | Quality Control (QC) & Safety Compliance – JCIA/DOH
Healthcare Equipment Lifecycle Management | Excel | Power BI | Python (Basic) | C++ (Basic)

PROFESSIONAL EXPERIENCE

Biomedical Sales & Service Trainee

April 2026 – May 2026

Hospitaall Needs – Calibration, Installation & Services | Coimbatore, India

- Gained hands-on exposure to medical equipment sales, installation, troubleshooting, preventive maintenance (PPM), and after-sales customer support in a clinical field service environment.
- Supported biomedical service engineers during on-site equipment installation and commissioning activities at hospital and clinic facilities.
- Performed preventive maintenance (PPM) checks and assisted with corrective maintenance (CMM) tasks on biomedical devices, ensuring compliance with safety and performance standards.
- Developed fault diagnosis, spare parts coordination, and service documentation skills across diverse equipment categories.

Biomedical Engineering Intern

June 2025 – June 2025

G. Kuppaswamy Naidu Memorial (GKNM) Hospital, Coimbatore | NABH-Accredited Tertiary Care Hospital

- Assisted in preventive maintenance (PPM) and corrective maintenance (CMM) of biomedical equipment across ICU, OT, and general wards.
- Supported calibration activities and equipment performance verification for patient monitors, ventilators, infusion pumps, ECG machines, and diagnostic imaging systems.
- Observed and documented equipment installation, commissioning, and acceptance testing procedures conducted by senior biomedical engineers.
- Performed equipment fault diagnosis, troubleshooting, and escalation procedures in line with hospital biomedical department protocols.
- Maintained accurate service records, maintenance logs, and technical documentation in compliance with hospital quality standards.

MEDICAL EQUIPMENT FAMILIARITY

Patient Monitors | Ventilators | Infusion Pumps | Syringe Pumps | ECG Machines | Defibrillators
Suction Pumps | Diathermy / Electrosurgical Units | Radiant Heat Warmers (Neonatal) | Infant Incubators | Clinical Beds

ACADEMIC PROJECTS

EMG-Controlled Game-Based Rehabilitation Therapy System | Design Patent Holder

- Designed an EMG-based control system for game-driven rehabilitation therapy targeting stroke recovery and neurological disorders including cerebral palsy.

- Developed biomedical signal acquisition and processing pipeline for real-time EMG signal interpretation and assistive device control.
- Design Patent granted by the Intellectual Property Office, Government of India — Design No. 491850-001, Class 24-02, dated 20/02/2026.
- Published co-authored research: EMG-Based Game Therapy for Stroke and Cerebral Palsy Rehabilitation (ISBN: 978-93-342-4986-6).

Wireless PPG-Based IoT Blood Pressure Monitoring System

- Designed a non-invasive blood pressure monitoring system using PPG sensor data with IoT-based wireless transmission and physiological data interpretation.
- Published co-authored research: Wearable Non-Invasive Optical Monitoring for Bladder Dysfunction and Enuresis (DOI: 10.63001/tbs.2024.v19.i03.pp120-125).

Bone Anchored Hearing Aid Prototype

- Developed a low-cost biomedical prototype leveraging bone conduction technology for hearing rehabilitation, contributing to hardware integration and clinical prototype testing.

MedVault – Medical Record Management System

- Designed a secure digital healthcare data management system for structured medical record handling and improved clinical data accessibility.

EDUCATION

B.E. Biomedical Engineering — Minor in AI/ML

2022 – 2026

Hindusthan College of Engineering and Technology (Anna University Affiliated), Coimbatore, India

CGPA: 8.37 / 10.0

PUBLICATIONS & PATENTS

- Wearable Non-Invasive Optical Monitoring for Bladder Dysfunction and Enuresis — DOI: 10.63001/tbs.2024.v19.i03.pp120-125
- EMG-Based Game Therapy for Stroke and Cerebral Palsy Rehabilitation — ISBN: 978-93-342-4986-6
- Design Patent: Travel-Friendly Portable BiPAP Device — Design No. 491850-001, IP India, Govt. of India (2026)

ADDITIONAL INFORMATION

Nationality: Indian

Current Location: Dubai, UAE

Availability: Immediate

UAE Driving License: Indian License — Eligible for Direct UAE Conversion (No Retest Required)

Languages: English, Tamil, Malayalam

Target Roles: Biomedical Technician | Biomedical Engineer | Field Service Engineer | Hospital BME Support